

TANIMA HOSSAIN

Software Engineer | AI
Research Engineer

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📍 Dhaka, Bangladesh

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</> Skills

Python, TypeScript, Rust, C/C++,
JavaScript, C#, Java, Express.js, Node.js,
React.js, Next.js, .NET 6, MySQL,
PostgreSQL, Sequelize ORM, Knex.js,
HTML, CSS, TailwindCSS, Git, Docker,
Postman, Playwright, Sanity.io, Stripe.

🎓 Education

BSc in Computer Science & Engineering

Daffodil International University ✨
01/2018 – 01/2022 | Dhaka, Bangladesh

🏆 Competitive Programming

Problem Solving

Solved over 3500 data structures and
algorithmic problems in various **online
judges** ✨.

Participated in numerous onsite and
online programming contests -

Individual Contest

- **Topper** ✨ in DIU Programmers
Individual Rank list - 2020 & 2021
- **Candidate Master** at Codeforces ✨,
Max Rating: 1974 | Best Ranks: Div. 1
✨ : 1028/10630 (Top 10%), Div. 2 ✨ :
83/10990 (Top 1%), Div. 3 ✨ :
197/4974 (Top 4%).
- **Knight** at LeetCode ✨, Max Rating:
1913 (Top 5%) | Best Rank: biweekly 93
✨ : 257/17080 (Top 2%).
- **4 star** at Codechef ✨, Max Rating:
1925 | Best Rank: Div. 2 ✨ : 44/5465
(Top 1%).
- **S ranked** at Toph ✨, Max Rating: 1879
| Best rank: 16th December ✨ : 9/321
(Top 3%).
- **Google CodeJam** 2020 | Round 1B,
Rank: 1392 | Round 2, Rank: 3291

Software Engineer and AI Research Engineer with 4+ years of experience
building production-grade **AI agent workflows**, orchestration systems, and
AI-assisted engineering tools. Experienced in developing agent platforms,
integrating LLMs and VectorDBs, improving workflow reliability, and
evaluating AI-generated outputs with a focus on practical business impact.

📁 Work Experience

Software Engineer

02/2026 – Present

Optimizely ✨

Dhaka, Bangladesh

- Engineering features and integrations for **Opal** ✨, Optimizely's no-code **AI agent workflow** orchestration platform.
- Optimised the agent ZIP import/export flow time by **~10x** using parallelised transfers and signed-URL direct uploads.
- Developed a versioning system for the Opal agent workflows, allowing checkpoints and rollback capabilities.
- Resolved **10+ production issues** in the agent orchestration service affecting agent lifecycle and data reliability.

AI Research Engineer

03/2025 – 02/2026

InfinitiBit Ltd ✨

Dhaka, Bangladesh

- Engineered the first Rust-based open source **AI agent framework**, **GraphBit** ✨, with a Python wrapper for speed and efficiency.
- Implemented support for multiple LLM providers for Graphbit, such as AzureOpenAI, MistralAI, Anthropic, and ReplicateAI.
- Integrated multiple VectorDBs such as ChromaDB, FAISS, Weaviate, ElasticSearch, Pinecone, Milvus, MariaDB, Qdrant with Graphbit.
- Developed multiple **AI agents with tool-calling** ability using Graphbit, such as a chatbot and resume ranker.
- Implementing a **cloud platform** for GraphBit, where people can publish and deploy AI agents as a service.

Software Engineer

02/2022 – 08/2024

Cefalo Bangladesh Ltd. ✨

Dhaka, Bangladesh

- Worked with a team to build the best web-based gamified maths learning platform, the **House of Math** ✨.
- Enhanced user experience by designing a **Graph Algorithm** for personalizing learning paths over 150+ topics.
- Minimised business impact by resolving over 100 **production issues**, ensuring smooth operation.
- Ensured comprehensive feature coverage by writing 30+ **End-to-End** and **Unit** tests using **Playwright** and Jest.

🏆 Honours and Awards

Champion, DIU Intra University Girls Programming Contest, 2019 ✨

Onsite Individual

I was awarded by Mr. Shahriar Manzoor, judge of ACM-ICPC World Finals, in the contest organized by Daffodil International University.

Champion, DIU Inter Department Math Olympiad 2019 ✨

Onsite Individual

Awarded the champion's trophy by the General Secretary of the Bangladesh Math Olympiad Committee for being the top scorer among 300+ participants.

Runner-up, DIU Intra University Programming Contest 2021 ✨

Onsite Individual

Runner-up, Inter University Girls' Programming Contest Season 2 ✨

Onsite Team contest

Participated as DIU_Heuristic, achieving Runner-up in the contest organized by North South University.